

Project Proposal: Evaluating and Enhancing Vision Model Robustness in Adverse Weather Conditions

Course: Laborprojekt Servicerobotik / Bildverarbeitung für Robotik (SS 2026)

Team Size: 3 Students

Estimated Workload: ~140 Hours (Per Team Member Effort)

Project Type: Software / Simulation-based Computer Vision

1. Motivation and Objectives

In autonomous driving and outdoor mobile robotics, robust visual perception is highly critical for safety. However, state-of-the-art object detection models often suffer severe performance degradation in adverse weather conditions (e.g., fog, heavy rain).

The primary objective of this project is to quantitatively evaluate this vulnerability and explore intervention strategies. Rather than building a system from scratch, this project focuses on **systematic evaluation, image preprocessing interventions, cross-architecture comparison, and Explainable AI (XAI)**.

2. Proposed Methodology

To ensure a reliable engineering pipeline and deep scientific analysis, the project will be executed using the following technology stack:

- **Datasets:** Open-source autonomous driving datasets with adverse weather subsets (e.g., Foggy Cityscapes or BDD100K).
- **Baseline Model (CNN):** YOLOv8 (Pre-trained) will serve as the primary baseline to establish the performance drop (mAP) caused by weather artifacts.
- **Preprocessing Interventions:** We will implement and test image enhancement algorithms (e.g., Dark Channel Prior for dehazing, CLAHE) as a pre-processing step to recover lost detection accuracy without retraining the models.
- **Cross-Architecture Extension:** To increase the academic depth, we will deploy a Transformer-based model (**RT-DETR**) alongside YOLOv8 to compare how different network architectures handle spatial noise and occlusions.
- **Explainable AI (XAI):** We will utilize **Grad-CAM** to generate attention heatmaps. This will allow us to qualitatively analyze *why* models fail in the fog and *how* preprocessing algorithms help refocus the network's attention on critical features (e.g., traffic signs, lane markings).

3. Project Scope & Risk Management

We have designed the project with a strict **"Core + Extension"** structure to guarantee deliverability:

- **Core Task (Baseline & Preprocessing, ~120h):** Ensures the basic pipeline (Data -> Inference -> Evaluation -> Enhancement -> Re-evaluation) is fully functional for a guaranteed "Pass" grade.
- **Advanced Extension (+50h):** The RT-DETR comparison and Grad-CAM

visualizations are modular add-ons that will elevate the quality of the final report without risking the core pipeline.

4. Timeline and Milestones (14 Weeks)

Phase 1: Baseline Establishment & Midterm (Weeks 1 - 5)

- Extract "Clear Weather" and "Adverse Weather" sub-datasets.
- Deploy YOLOv8 and RT-DETR pre-trained models.
- Run batch inference to calculate and plot the exact mAP performance drop.
- **Milestone 1:** Midterm Presentation demonstrating the verified vulnerability of the models.

Phase 2: Interventions & Cross-Evaluation (Weeks 6 - 9)

- Apply image enhancement algorithms (Dehaze/Derain) to the adverse weather datasets.
- Re-run inference on both YOLOv8 and RT-DETR using the enhanced images.
- Evaluate and compare the accuracy recovery rates between the CNN and Transformer architectures.

Phase 3: Deep Analysis & XAI (Weeks 10 - 12)

- Select representative failure and recovery cases.
- Generate Grad-CAM heatmaps for these specific cases to visualize the shift in the models' attention mechanism.
- Begin drafting the final report.

Phase 4: Finalization (Weeks 13 - 14)

- Clean, document, and publish the code repository.
- Finalize the LaTeX report (5-10 pages) containing quantitative charts and qualitative XAI heatmaps.
- **Milestone 2:** Final Presentation and Project Defense.

5. Expected Deliverables

1. **Midterm Presentation:** Problem statement and baseline degradation results.
2. **Open-source Codebase:** Reproducible pipeline for evaluation, preprocessing, and XAI heatmap generation.
3. **Final Report (LaTeX, 5-10 pages):** Comprehensive analysis of the findings, including quantitative metrics (mAP comparisons) and qualitative proofs (Grad-CAM).
4. **Final Presentation.**

